

Fostering Data Worker Inclusion and Well-Being: Identifying Barriers and Designing Interventions

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ABSTRACT

In recent years, we have seen great advancements in artificial intelligence. Given this new “Age of AI”, an increase in annotated data for training advanced models requires increased support of a large, diverse, and robust population of workers. Fostering inclusivity and well-being within this space of data work must then be a top priority. In my thesis, I start with identifying barriers to inclusion and well-being across three groups of data workers with differing challenges. First, I study crowd workers whose disabilities encumber their ability to participate in data work. Next, I explore the effectiveness of a strategy for training crowd workers to complete complex tasks, as advanced AI models may elevate the required complexity of future data work. Lastly, I will briefly discuss a work-in-progress characterizing the productivity and well-being of a trained annotation team. Through these studies, I find a common shared barrier: a lack of shared knowledge of productive behaviors among workers. I conclude with future work involving designing interventions to mitigate this barrier.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

crowdsourcing; accessibility; goal setting; complex work; productivity

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1 INTRODUCTION

As artificial intelligence (AI) continues to receive widespread public attention, with Bill Gates dubbing this “The Age of AI” [6], many forget that the backbone of such innovation is human intelligence. We see humans participating in the labor of generating and annotating the data that AI models are trained on, known as “data work”, across a spectrum of both complexity of the labor and tactics

of organization. On one side of the spectrum we have microtask crowdsourcing work, which involves rapid mobilization of a large, distributed crowd of workers performing small tasks that require basic human intelligence to complete. On the other side, trained in-house annotation workforces are recruited to perform more complex labeling tasks.

Though the requirements for participating vary, we must still consider inclusion and well-being within this data work economy. As the demand for annotated data rises, an increase in data workers will be necessary, and for greater retention and productivity of these workers, prioritizing their well-being is imperative [13, 20]. However, a larger workforce alone is not enough. For one, datasets are frequently created with built-in bias [21] or lacking representation of people of varying races or cultures [1], something that can be mitigated by incorporating a diverse set of data workers. We may also soon see a shift in the needed complexity of data work as AI’s ability to perform “human intelligence” tasks matures [3], excluding those that cannot adapt to more advanced work.

Taking inspiration from Maslow’s Hierarchy of Needs [11], we can imagine a hierarchy of needs for inclusion and well-being of data workers. At its base, I consider the need for accessibility to full participation within the current data work economy, a need that can be unmet by those who struggle due to disability [24] or lack of reliable internet access [5]. Beyond that, I recognize the need for the ability to perform complex work, providing continued inclusion within the data work economy. Finally, once underlying needs are met, I envision the need for data workers already trained in complex work to reach their full potential through greater productivity and well-being.

In my thesis, I start with identifying barriers to inclusion and well-being across three groups of data workers with differing challenges, looking for common barriers to mitigate through specialized intervention. First, I survey crowd workers with disabilities to understand their experiences performing data work [15]. Next, I explore the effectiveness of goal setting as a strategy for training crowd workers to complete complex tasks [14]. Lastly, I briefly discuss a work-in-progress characterizing the productivity and well-being of trained annotators. One mutual barrier found is a lack of shared knowledge of productive behaviors among workers, especially those with similar personal characteristics or task expectations. I conclude with future work on designing interventions to recommend actions to data workers based on productive behaviors of workers akin to them.

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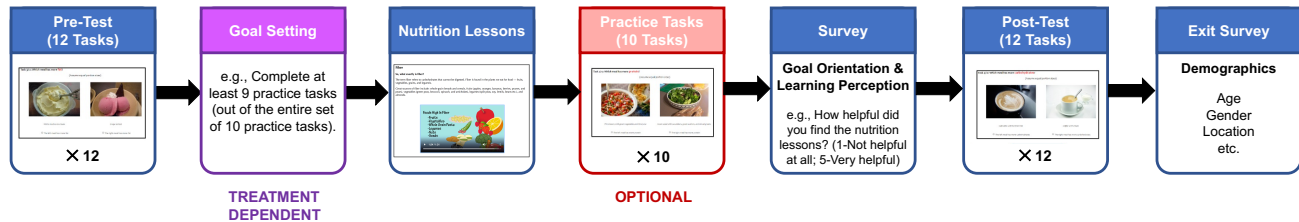


Figure 1: An overview of our Phase 1 procedure.

2 UNDERSTANDING CHALLENGES FOR CROWD WORKERS WITH DISABILITIES

Beyond the benefit that the diversity of crowd workers with disability could provide to the data work economy, crowd work holds great opportunity for workers with disability as well. Such benefits as the ability to work from home and at one's own pace, as well as the anonymity of crowd platforms can offer workers with disability greater comfort, flexibility, and ease of mind. Despite these benefits, most of the literature exploring the space of the accessibility of crowdsourcing, most notably the Amazon Mechanical Turk (MTurk) platform, find several issues, including tasks and platform features that are inaccessible to those with a variety of conditions [2, 19, 24] and infeasible expectations for work completion [22, 24]. However, much of the prior work focuses only on workers with disabilities that directly affect computer usage, rather than also including mental health concerns and Autism Spectrum Disorder, and there is no comparison of the extent of the issues that workers with and without disability face as a juxtaposition.

To fill in this gap, I engaged in a two-phase survey study to explore the following: 1) how workers with and without disability differ in their experiences using MTurk, 2) how disability impacts wages earned through MTurk, and 3) the nature of interactions with other crowd workers for workers with disability. Our first survey was open to all MTurk workers and included topics such as usability of the platform, wages earned, interactions with others, and disability status. Based on our findings from the first survey, I developed our second survey to probe deeper into these topics and invited back our survey respondents who self-identified as having a disability.

I find that workers with disability experience greater difficulty with using the MTurk platform, including difficulty finding a task that they can work on and completing tasks within the allotted time. These difficulties can occur and impact wages when a disability hinders one's working speed, requires greater amounts of rest, or causes an incompatibility with certain tasks (especially when this incompatibility is not immediately obvious from a task description). Though receiving adequate pay is a common issue for most crowd workers [10], low earnings disproportionately impact those with disability given our finding that workers with disability are significantly more likely to have MTurk be their full-time source of income. Workers with disability are also significantly more likely to take part in online interactions with other workers and to be

actively engaged in these online interactions, though these interactions are often limited to assistance for finding worthwhile tasks, with many workers deliberately choosing to hide their disability status.

Overall, I find several barriers to full inclusion and well-being of workers with disability, including disproportionately inadequate income due to issues such as inability to finish tasks on time and incompatibilities between one's condition and tasks. Though issues such as low pay and tasks that do not provide enough time would require platform support to regulate a set of standards in order to solve, we can imagine a solution that can help workers find tasks compatible with their condition as a first step towards improving inclusion and well-being.

3 FACILITATING COMPLEX TASK TRAINING THROUGH GOAL SETTING

Many approaches have been developed to enable the completion of complex tasks by the crowd with training being one popular method, including training by examples and gold standard tasks [9, 12] and training by self-evaluation, expert assessment, or peer feedback [4, 23]. Taking a page out of the fields of educational and organizational studies, prior work has found that goal setting during training can be a successful motivational strategy for enhancing learning and performance, with its effectiveness depending on both who sets the goal and the type of goal [8, 17]. Though goal setting has been found to have merit in traditional learning environments, its effects on learning and performance when training crowd workers to complete complex tasks remains unclear.

To answer this question, I conducted a three-phase experimental study to explore the following: 1) the impact of goal setting on workers' learning perceptions, and 2) the impact of goal setting on workers' learning gain during training and performance after training. A workflow of the first phase of our study can be found in Figure 1. I evaluated workers on their knowledge of a complex task both before and after going through a goal setting procedure, a set of relevant lessons, and a series of optional practice tasks, with the difference in these evaluations representing their learning gain. During goal setting, workers were divided into treatments based on who set their goal and the type of goal. After the lessons, workers were surveyed on their perceptions of the learning material and their personal attitudes towards achievement. Workers were further evaluated days afterward on the complex task in phases two and three to evaluate post-training performance.

Goal setting was found to have a significant effect on workers' perceptions of learning, but not their learning gain during training or their post-training performance. Despite these negative findings, engaging in more of a desirable behavior for improving learning (i.e., the practice tasks) was found to be associated with both improved learning gain and post-training performance, but only when the worker chose to engage in the behavior themselves rather than being assigned to perform the behavior as a goal. Lastly, personalization of goals towards worker attitudes also showed promise towards improving learning gain.

The findings of my study differ significantly from prior literature on goal setting, suggesting that differences in typical educational environments and the crowdsourcing environment may be a large barrier towards greater effectiveness of goals. Though certain factors of crowd work such as its anonymity and remoteness cannot be changed, allowing workers to assist one another in their learning can overcome the typically independent nature of this work and allow for the encouragement of desirable learning behaviors.

4 CHARACTERIZING ANNOTATOR PRODUCTIVITY AND WELL-BEING

Though the future inclusion of annotators trained to engage in complex work is more secure than the populations of data workers previously discussed, there is the question of whether workers can ultimately reach their full potential without taking steps to ensure greater productivity and well-being of the work. Certainly, we can imagine that failing to take these factors into account will impact attrition within the data work economy. A look into the productivity of data workers begins with a discussion of how we define productivity, something that is not always obvious within the space of knowledge work [7], and also how markers of productivity change based on both characteristics of the task and personal characteristics of the worker. As complex labeling tasks become more prevalent for work in high-impact areas such as autonomous vehicle training [16], identifying what productive behavior looks like, as well as which factors are most likely to cause worker well-being to suffer during task completion, will become more important.

In this current work-in-progress, I conducted an experimental study that recruited trained annotators from a large technology company to work on a real-world complex labeling task. Variations of the instructions of this task were given to participants as characteristics along two dimensions: 1) number of constraints when determining whether or not an annotation should be created, and 2) level of granularity when determining an appropriate label for an annotation. Telemetry recording participants' actions while working on the task was collected, and afterward participants were surveyed about their difficulty and workload [18] experienced during the task.

As preliminary findings, we see indications that characteristics of a given task can impact productivity in terms of label quantity, but additional work will need to be done to effectively measure label quality in order to reliably determine its role towards characterizing productivity. We also see indications that factors like prior experience an annotator has with the type of task as well as characteristics of the task can influence perceived difficulty and

workload, with factors such as a high number of constraints or high experience with the task increasing mental load.

Some initial barriers found to worker productivity and well-being include the changing nature of what defines productivity based on task characteristics, as well as the unintuitive nature of certain complex labeling requirements. Added mental workload can also quickly add perceived difficulty to such tasks. With this in mind, a solution that helps reduce mental load needed while remaining personalized to the task at hand may help alleviate these issues.

5 FUTURE WORK

Taking the identified barriers altogether (difficulty finding sufficient tasks to work on, lack of assistance from workers who engage in advantageous learning behaviors, and inconsistent patterns of productivity based on task), we can combine them to represent a lack of shared communication and knowledge among data workers. With this in mind, we can envision a system to help mitigate these issues. The core principle involves identifying data workers with greater amounts of knowledge in some capacity and recommending their behaviors to other workers when appropriate, specifically to workers akin to them in terms of either personal or task characteristics.

I am interested in starting with a system that recommends tasks for workers to complete based on tasks completed by workers with similar capabilities, a solution that can assist all data workers, including those with disability. From there, a system to help data workers train to complete complex tasks can better segment training into a set of quantifiable behaviors, learning a pattern of productive behaviors from other workers that result in higher learning and performance to recommend to others. Lastly, we can imagine a similar system for assisting trained annotators, such that once a core idea of productivity can be defined, productive patterns of behavior can be measured with next actions recommended to those deviating from the expected path.

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